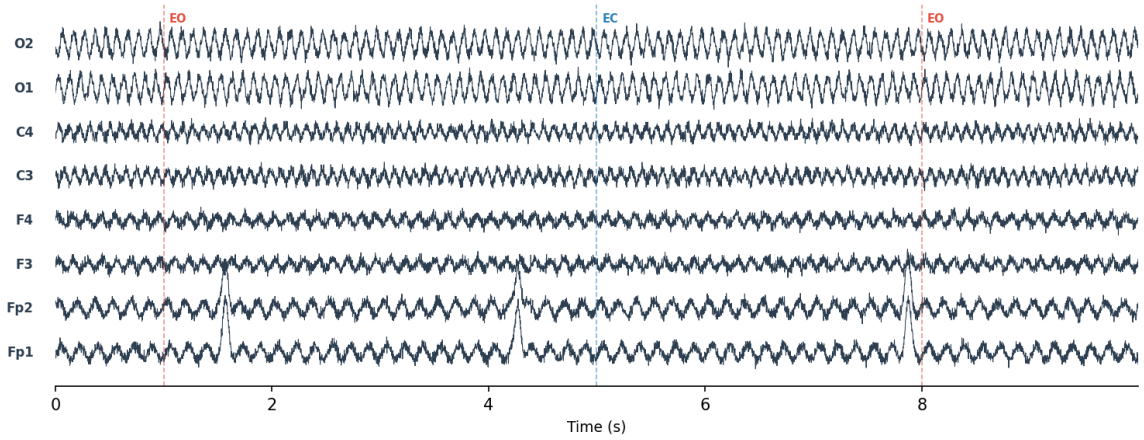
EEG Analysis Report

Spectral analysis, band powers, and functional EEG features

FILE: demo_recording.easy
DEVICE: ENOBIO CHANNELS: 8 SAMPLING RATE: 500 Hz
TOTAL DURATION: 10.0 s ANALYSIS WINDOW: 0.0 – 10.0 s (10.0 s)
DATE: April 09, 2026

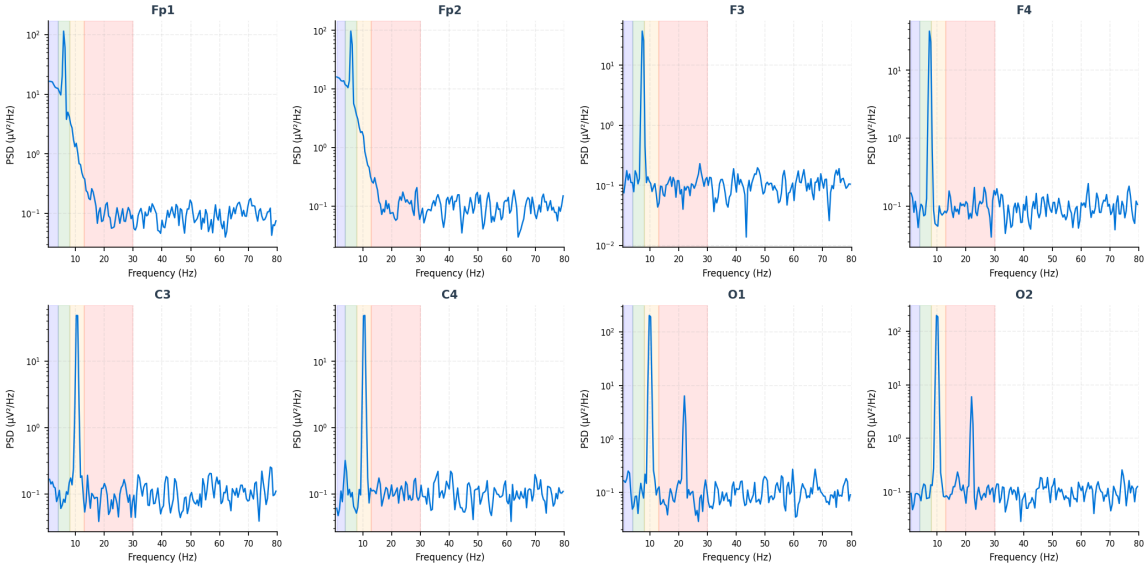
1. Raw EEG Traces



Markers detected: EO @ 1.00s, EC @ 5.00s, EO @ 8.00s

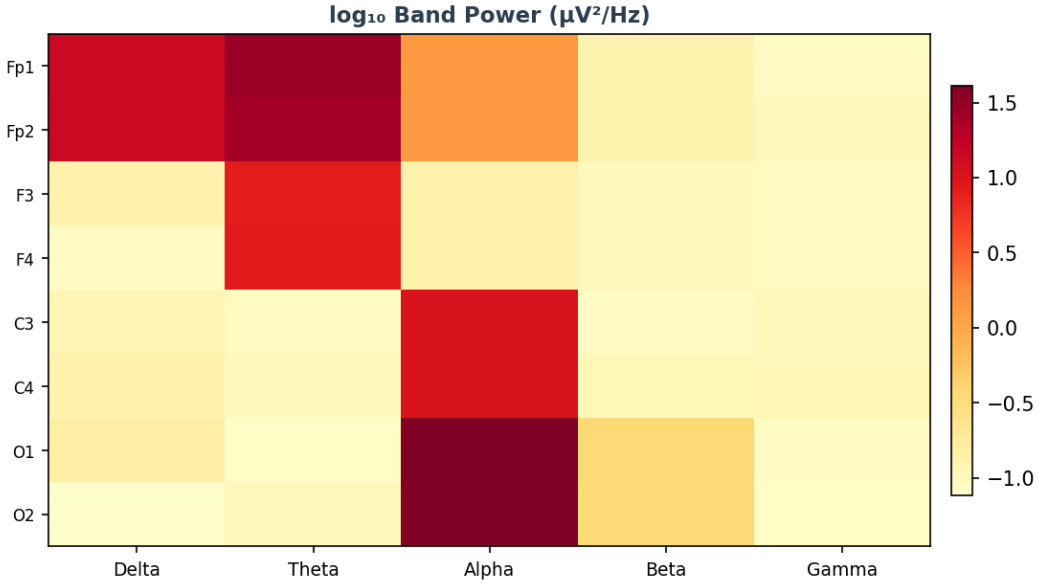
2. Power Spectral Density — Channels 1–8 (Fp1, Fp2, F3, F4, C3, C4, O1, O2)

Welch's method, 1024-sample segments, computed over full 10.0 s analysis window.



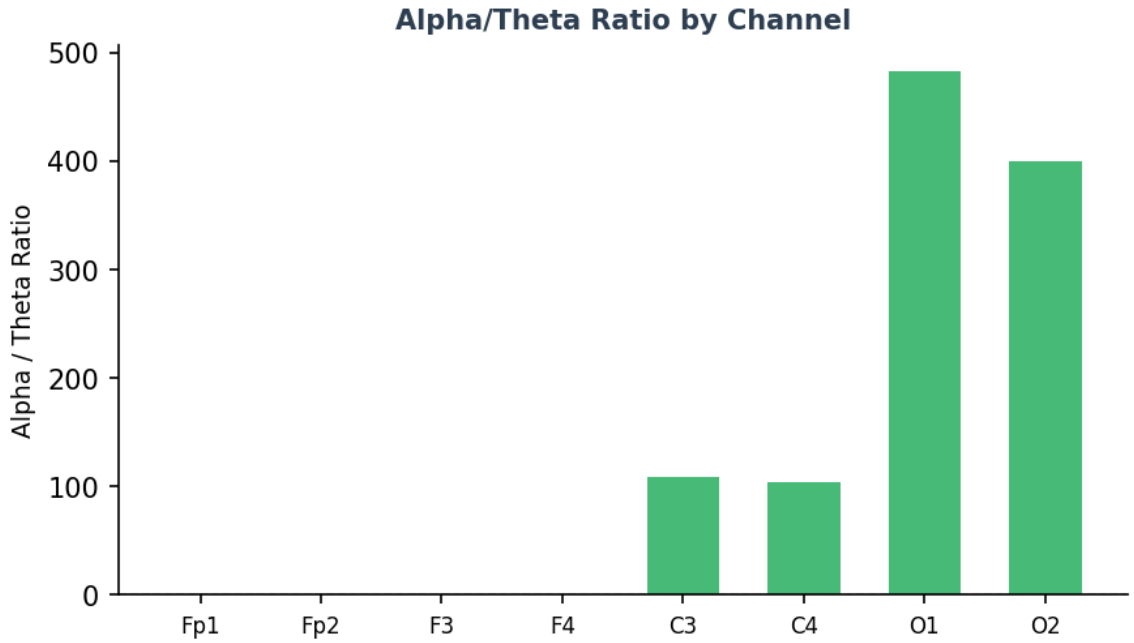
3. Spectral Overview

Band power distribution across all 8 channels. Values are log■■(mean PSD) in each standard frequency band, computed over the full 10.0 s analysis window.

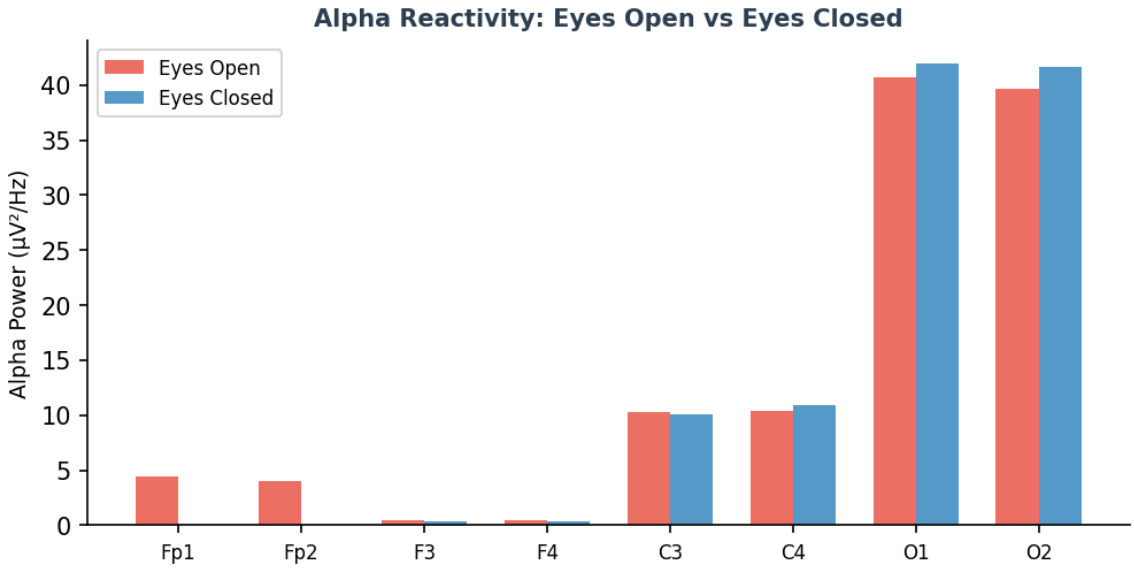


4. Alpha Analysis

Alpha/theta power ratio by channel. Ratio > 1 indicates alpha-dominant activity (green), ratio < 1 indicates theta-dominant (orange). Posterior alpha dominance is a marker of normal resting-state EEG.



Alpha reactivity: comparison of alpha-band (8–13 Hz) power during Eyes Open vs Eyes Closed segments. Healthy subjects typically show increased posterior alpha during EC (Berger effect).



Frontal Alpha Asymmetry (ln F4 – ln F3): 0.0335 → right > left (approach tendency). Positive values indicate relatively greater left frontal activity.

5. Spectral Summary

Mean PSD (µV²/Hz) per frequency band, alpha/theta ratio, and peak alpha frequency for each channel. Computed over the full 10.0 s analysis window.

Channel	Delta	Theta	Alpha	Beta	Gamma	α/θ	Peak α (Hz)
Fp1	14.162	28.529	1.372	0.126	0.091	0.05	8.3
Fp2	13.968	25.444	1.329	0.127	0.101	0.05	8.3
F3	0.132	8.180	0.133	0.105	0.094	0.02	8.3
F4	0.090	8.416	0.137	0.104	0.095	0.02	8.3
C3	0.118	0.094	10.202	0.093	0.107	108.53	10.3
C4	0.129	0.100	10.342	0.110	0.112	103.73	10.7
O1	0.159	0.085	41.167	0.363	0.092	482.90	9.8
O2	0.077	0.101	40.293	0.354	0.085	400.30	9.8

6. Per-Channel Statistics

Descriptive statistics (µV) for each channel over the analysis window.

Channel	Mean (µV)	Std (µV)	Min (µV)	Max (µV)	Pk-Pk (µV)
Fp1	1.5	13.9	-28.2	98.1	126.3
Fp2	1.6	13.5	-26.6	102.0	128.6
F3	-0.0	7.5	-22.7	22.2	44.9
F4	0.0	7.6	-25.7	22.7	48.3
C3	0.0	8.7	-25.0	24.1	49.0
C4	0.1	8.7	-23.6	24.9	48.5
O1	0.0	15.2	-36.8	36.1	72.8
O2	0.0	15.1	-39.9	38.5	78.4